Università degli studi di Padova

Stochastic Methods for Finance

Report 3: Convergence of Call prices with binomial model

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1 Introduction

This report analyzes the pricing of European call options using three approaches: the Black–Scholes model, the Cox–Ross–Rubinstein (CRR) binomial lattice, and the Leisen–Reimer (LR) correction. The focus is on comparing numerical accuracy and convergence toward the Black–Scholes benchmark.

2 Black–Scholes Model

The Black–Scholes model provides a continuous-time framework for pricing European options and serves as the benchmark in this study.

2.1 Assumptions

The model assumes:

- Geometric Brownian motion for the underlying asset
- Constant volatility σ and interest rate r
- Frictionless markets with no arbitrage
- European exercise style

Under the risk-neutral measure:

$$dS_t = rS_t dt + \sigma S_t dW_t^Q$$

2.2 Pricing formula

The European call price is:

$$C_0 = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2)$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

This value is used as the benchmark for evaluating numerical methods.

3 Cox–Ross–Rubinstein Binomial Model

The CRR model approximates the evolution of the asset price using a recombining binomial tree over n time steps of size $\Delta t = T/n$.

3.1 Tree dynamics

At each step:

$$S \to \begin{cases} Su \\ Sd \end{cases} \qquad u = e^{\sigma\sqrt{\Delta t}} \quad d = e^{-\sigma\sqrt{\Delta t}}$$

The risk-neutral probability is:

$$q = \frac{e^{r\Delta t} - d}{u - d}$$

3.2 Pricing

The option is valued by backward induction:

$$V_t = e^{-r\Delta t}\left(qV_{t+1}^{(u)} + (1-q)V_{t+1}^{(d)}\right)$$

with terminal payoff:

$$V_T = \max(S_T - K, 0)$$

The model converges to Black–Scholes as $n \to \infty$, but exhibits slow and oscillatory convergence.

4 Leisen–Reimer Binomial Model

The Leisen–Reimer method improves the CRR model by constructing a binomial tree that better matches the Black–Scholes terminal distribution, leading to faster convergence.

4.1 Probability construction

Define:

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} \qquad d_2 = d_1 - \sigma\sqrt{T}$$

The transition probabilities are obtained using a refined inverse normal approximation:

$$p = \Phi_{PP}(d_2) \qquad \tilde{p} = \Phi_{PP}(d_1)$$

4.2 Tree construction

Given $\Delta t = T/n$, the parameters are:

$$u = e^{r\Delta t}\frac{\tilde{p}}{p} \qquad d = \frac{e^{r\Delta t} - pu}{1-p}$$

ensuring the martingale condition:

$$e^{r\Delta t} = pu + (1-p)d$$

A practical implementation detail is that an odd number of steps n is typically preferred to improve symmetry around the strike region and reduce discretization bias.

4.3 Pricing

Option values are computed by backward induction:

$$V_t = e^{-r\Delta t}\left(pV_{t+1}^{(u)} + (1-p)V_{t+1}^{(d)}\right) \qquad V_T = \max(S_T - K, 0)$$

The method reduces oscillations and achieves faster convergence toward the Black–Scholes benchmark compared to CRR.

5 Numerical Implementation

The numerical methods used in this report were implemented in VBA. The implementations are based on existing reference code, adapted to the specific option pricing framework used in this study.

5.1 Black–Scholes model

The Black–Scholes formula was implemented in VBA based on the reference code provided by Anthony VBA Finance:

- http://www.anthony-vba.kefra.com/vba/vba6.htm

5.2 Cox–Ross–Rubinstein model

The CRR binomial lattice implementation follows the structure provided in the same reference collection:

- http://www.anthony-vba.kefra.com/vba/vba7.htm

5.3 Leisen–Reimer model

The Leisen–Reimer implementation is based on the framework described in Vinegar Hill Finance Labs:

- https://sites.google.com/view/vinegarhill-financelabs/binomial-lattice-framework/leisen-reimer

6 Numerical Results

The numerical analysis is performed using the following parameters:

$$S_0 = 100 \quad K = 100 \quad r = 0.01 \quad \sigma = 0.2 \quad T = 1$$

The Black–Scholes price is used as the benchmark reference value for all comparisons.

6.1 Option prices and convergence

Table 1 reports representative values of the binomial CRR model and the Leisen–Reimer model compared to the Black–Scholes price for selected values of n.

n	CRR	BS	LR	$\lvert CRR - BS \rvert$	$\lvert LR - BS \rvert$
1	10.41	8.43	8.27	1.981	0.168
10	8.35	8.43	8.43	0.083	0.003
50	8.44	8.43	8.43	0.006	0.000
100	8.44	8.43	8.43	0.010	0.000
500	8.44	8.43	8.43	0.004	0.000
1000	8.43	8.43	8.43	0.001	0.000

Table 1: Representative option prices and absolute errors for CRR and Leisen–Reimer models.

6.2 Price convergence

Figure 1 illustrates the convergence of both numerical methods toward the Black–Scholes benchmark as the number of time steps increases.

The CRR model exhibits oscillatory convergence, while the Leisen–Reimer method converges more smoothly and rapidly.

6.3 Error analysis

Figure 2 shows the absolute pricing error $|V_n - V_{BS}|$ on a logarithmic scale.

Both methods display a power-law decay of the form:

$$|V_n - V_{BS}| \sim n^{-\alpha}.$$

A linear regression in log-log scale is used to estimate the convergence rate.

6.4 Convergence results

The estimated convergence rates are:

$$\alpha_{CRR} \approx 0.98, \qquad \alpha_{LR} \approx 1.43.$$

The CRR model shows a convergence rate close to the theoretical $\mathcal{O}(n^{-1/2})$ behavior but improved due to the finite range of n considered. The Leisen–Reimer method significantly improves convergence speed, consistent with its theoretical advantages.

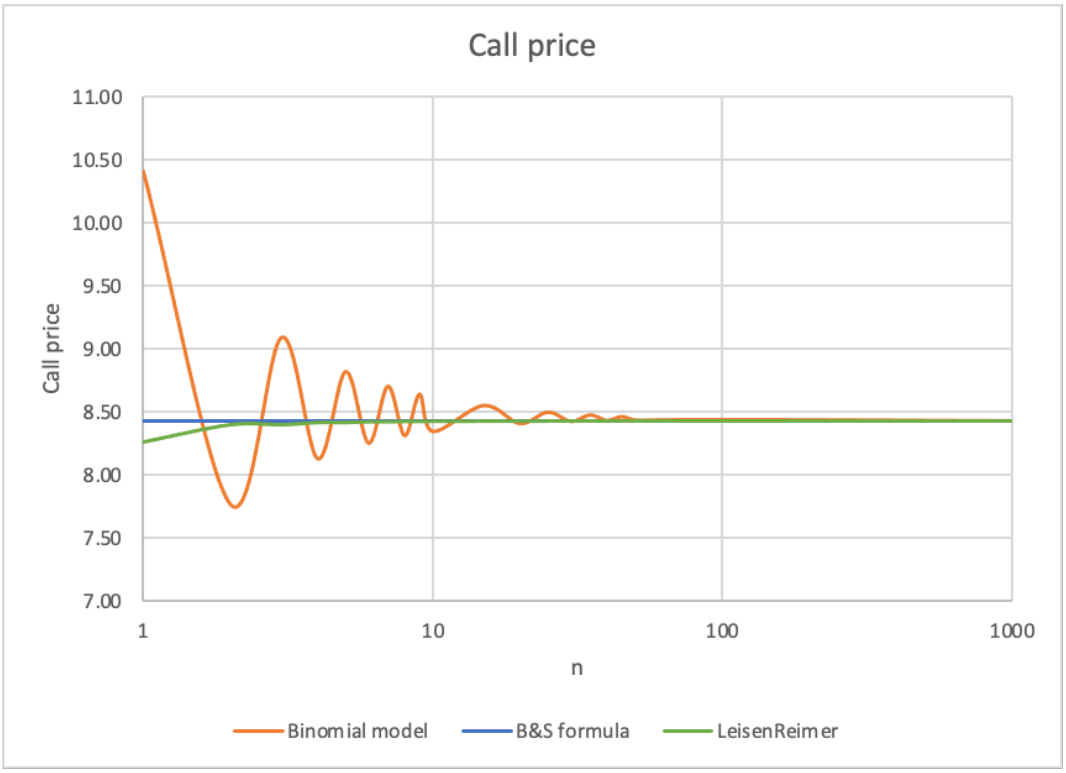

Figure 1: Option prices as a function of n.

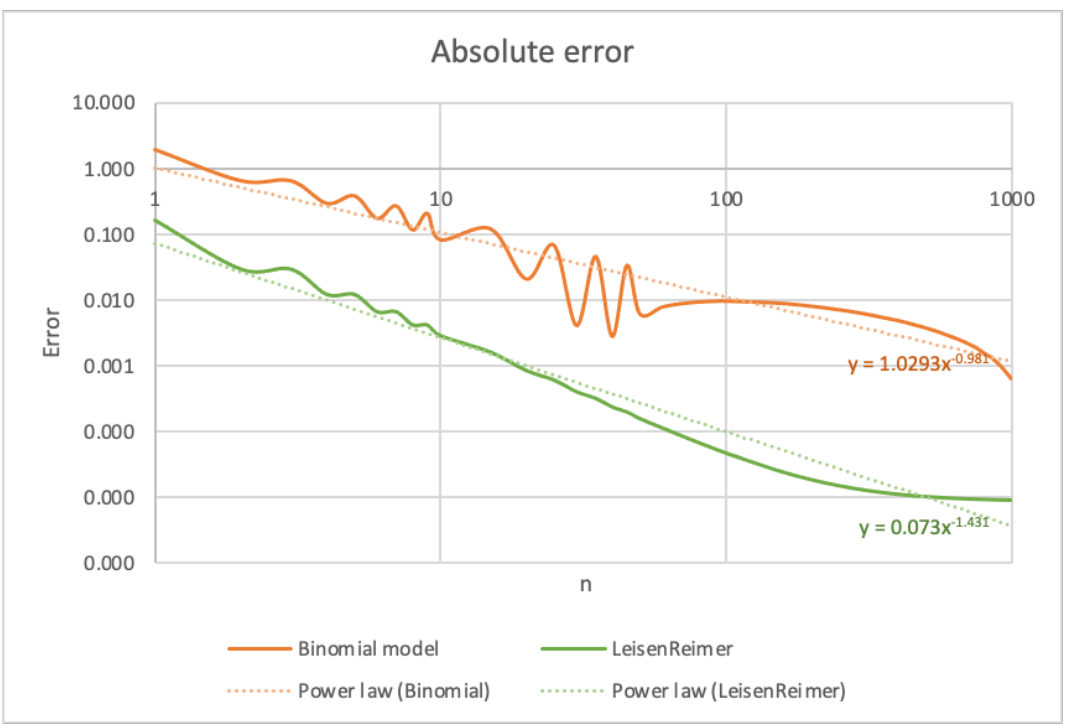

Figure 2: Absolute error $|V_n - V_{BS}|$ vs n (log scale) with fitted slopes.